Signals and Systems

Audio-Signal Transformations and Stereo Vocal Reduction

Course Project Report

Portfolio edition. This report removes personal identifiers, separates reproducible evidence from subjective listening, and adds a controlled ground-truth experiment for the vocal-reduction methods.

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October 2026

Abstract

This project applies core signals-and-systems concepts to digital audio. The first part demonstrates time reversal and scaling, the discrete Fourier transform (DFT), magnitude-only and phase-only reconstruction, and ideal low-pass filtering. The second part studies stereo centre cancellation as a simple vocal-reduction technique. In addition to the original $L - R$ baseline and a low-frequency "bass rescue" variant, this edition introduces a gain-and-delay compensated method for imperfectly centred vocals. A synthetic benchmark with known vocal and accompaniment components shows the central trade-off: under a two-sample, 0.82× channel mismatch, compensation improves vocal attenuation from −10.69 dB to −26.54 dB, while accompaniment quality falls to 12.06 dB SDR. The result is therefore an improvement for a specific failure mode, not a universal vocal separator.

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1 Objectives and implementation scope

The course specification asks for two projects. Project 1 covers five operations on a supplied audio signal: waveform display, time transformations, frequency spectra, reconstruction from magnitude or phase, and ideal low-pass filtering. Project 2 offers a choice of applications; this work implements the voice-eliminator option and discusses both theory and practical improvement.

All processing is implemented in MATLAB. The public-facing repository excludes the original songs, submission archives, student identifiers, and the course handout. The controlled benchmark creates its own signals and therefore remains fully reproducible without copyrighted audio.

2 Project 1: time- and frequency-domain processing

2.1 Time transformations

Let $f[n]$ be a sampled audio signal at sampling frequency F_s. Time reversal is implemented by reversing the sample order. For a positive scale factor a, the continuous-time relation $g(t) = f(at)$ is approximated on the original sampling grid using linear interpolation:

$$g[n] = f\left(a\frac{n}{F_s}\right). \tag{1}$$

Values $a = 3$ and $a = 1/3$ produce compression and expansion respectively. Figure 1 shows the resulting duration changes.

The interpolation is intentionally elementary. In a production resampler, compression should include an anti-aliasing filter before samples are removed. This limitation is retained and documented instead of being hidden by plotting.

2.2 Discrete Fourier spectra

For N samples, the DFT is

$$X[k] = \sum_{n=0}^{N-1} f[n]e^{-j2\pi kn/N}. \tag{2}$$

The implementation uses exact DFT-bin frequencies, spaced by F_s/N, and plots the normalized magnitude $|X[k]|/N$. This corrects the endpoint error introduced by using `linspace(-Fs/2, Fs/2, N)` for a DFT axis.

2.3 Magnitude and phase reconstruction

Writing $X[k] = |X[k]|e^{j\phi[k]}$, two deliberately incomplete inverse DFTs are formed:

$$f_{\text{mag}}[n] = \text{IDFT}\{|X[k]|\}, \tag{3}$$

$$f_{\text{phase}}[n] = \text{IDFT}\{e^{j\phi[k]}\}. \tag{4}$$

The magnitude-only result preserves energy distribution but loses temporal placement. The phase-only result has a very different amplitude scale yet can retain recognizable timing structure. Figure 3 therefore avoids claiming that either component alone reconstructs the original waveform.

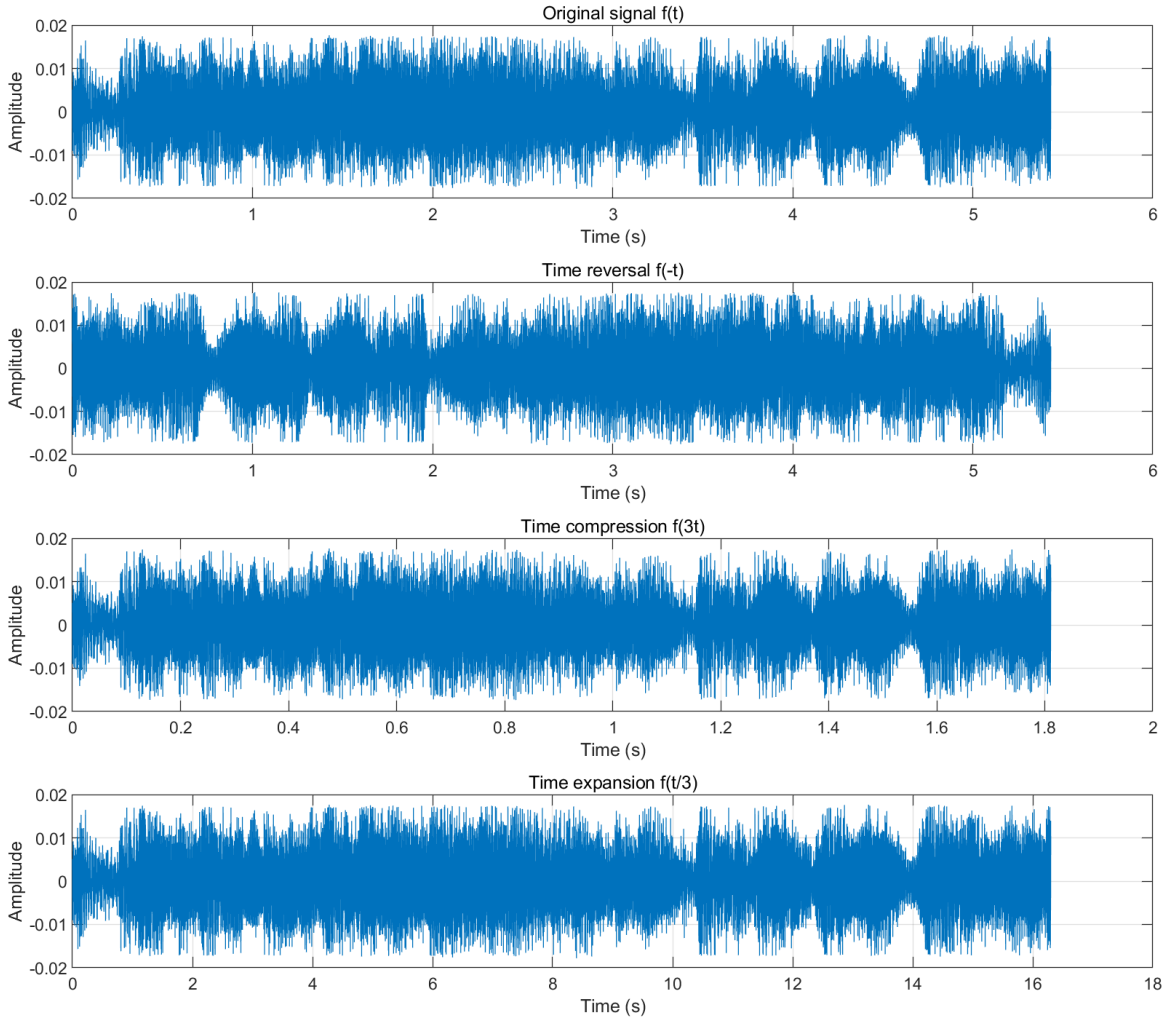


Figure 1: Original signal, time reversal, compression, and expansion. Each panel uses its own meaningful time span rather than forcing all plots onto the longest duration.

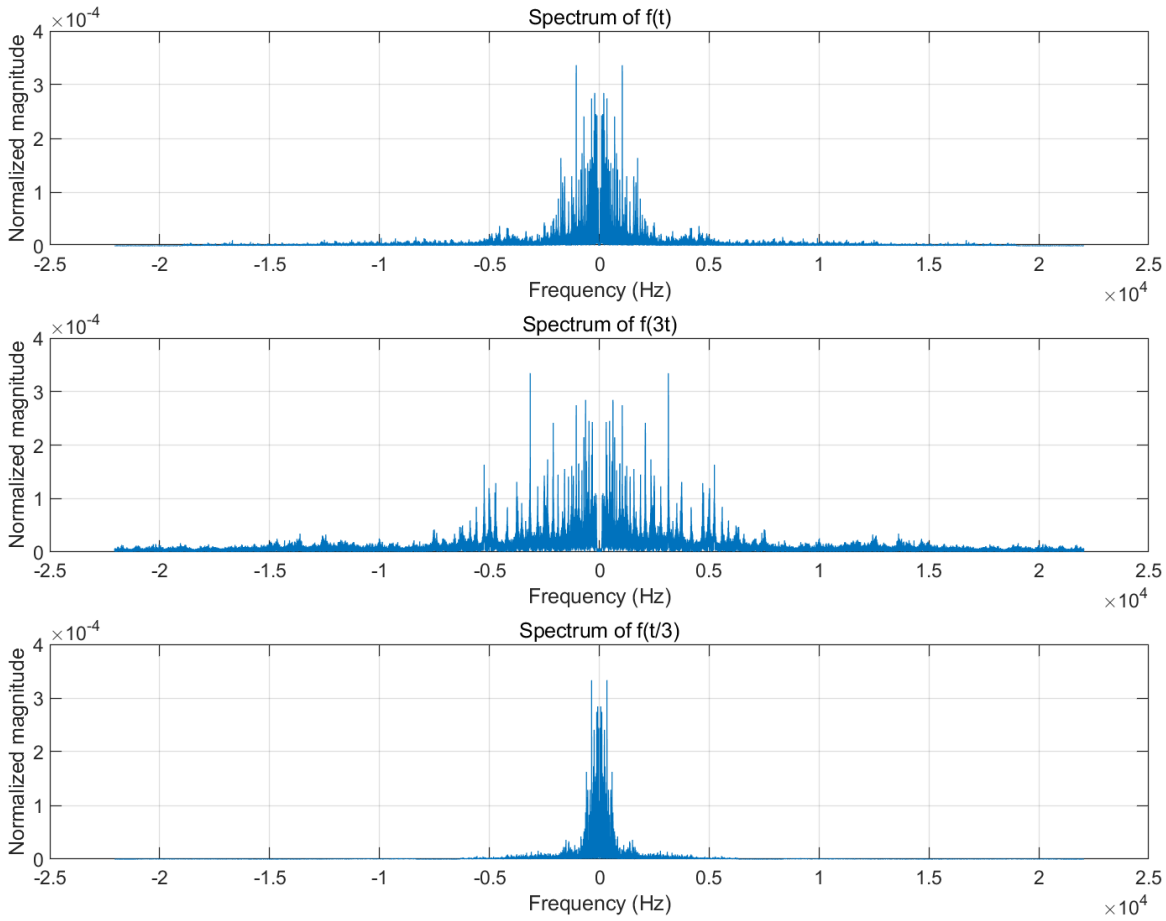


Figure 2: Normalized two-sided spectra for the original, compressed, and expanded signals. Time compression broadens spectral content; time expansion contracts it.

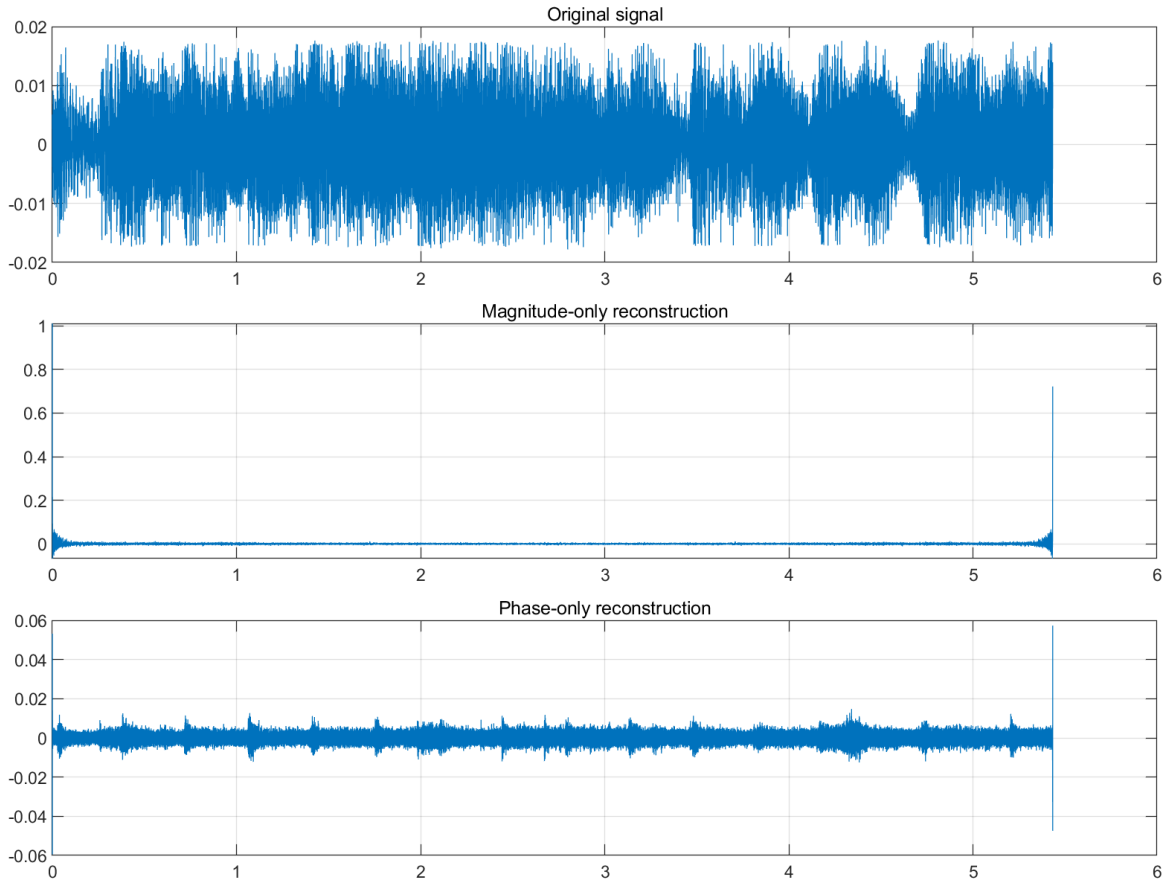


Figure 3: Original signal and inverse transforms using magnitude or phase information alone.

2.4 Ideal low-pass filtering

The ideal frequency response is

$$H(f) = \begin{cases} 1, & |f| \le f_c, \\ 0, & |f| > f_c, \end{cases} \qquad f_c = 1000 \text{ Hz}, \tag{5}$$

and the output is obtained from $y[n] = \text{IDFT}\{X[k]H[k]\}$.

The implementation directly masks a finite-length DFT. Consequently it has a brick-wall response, a long impulse response, circular boundary assumptions, and possible ringing. Those are properties of the idealized exercise, not a recommended real-time audio filter.

3 Project 2: stereo vocal reduction

3.1 Stereo model and basic cancellation

Assume a centre-panned vocal $v[n]$ and channel-dependent accompaniment $a_L[n], a_R[n]$:

$$L[n] = a_L[n] + v[n], \qquad R[n] = a_R[n] + v[n]. \tag{6}$$

Then

$$y_{\text{basic}}[n] = L[n] - R[n] = a_L[n] - a_R[n] \tag{7}$$

cancels an exactly equal centre source. It also cancels every other equal centre-panned element, including bass, kick, snare, and lead instruments.

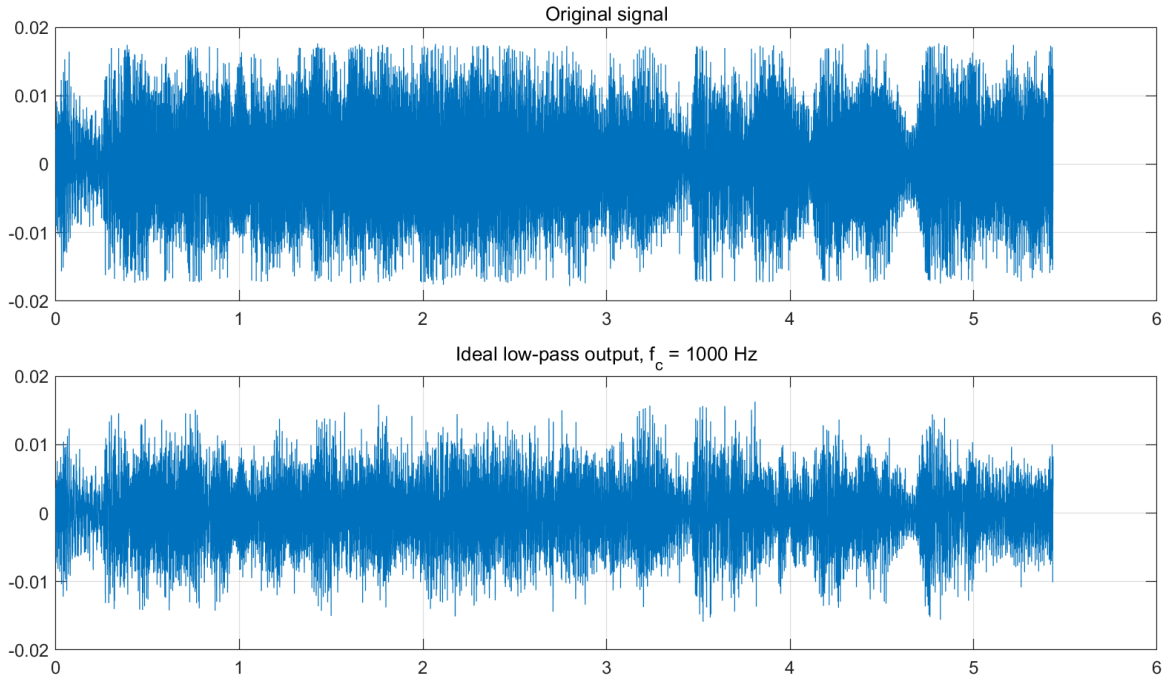


Figure 4: Original signal and ideal low-pass output at 1 kHz.

3.2 Legacy bass-rescue baseline

The original extension added a low-frequency part of the mid channel:

$$y_{\text{bass}}[n] = L[n] - R[n] + 0.30\ \text{LPF}_{160\,\text{Hz}}\left\{\frac{L[n]+R[n]}{2}\right\}. \tag{8}$$

This can restore bass energy, but it is not unconditionally "more advanced." Male vocal fundamentals can lie near or below the cutoff, so the method can also reintroduce vocal content. It is kept as a documented legacy baseline.

3.3 Gain-and-delay compensated cancellation

Real mixes may place the nominal centre source in the two channels with a small gain difference or sample delay. The improved method first isolates a broad speech band (120–6000 Hz) for parameter estimation. It searches integer delays $d \in [-8, 8]$ and maximizes normalized correlation:

$$\hat{d} = \arg\max_{d} \frac{\langle L_b,\, R_b(\cdot + d)\rangle}{\|L_b\|_2\, \|R_b(\cdot + d)\|_2}. \tag{9}$$

After alignment, an energy-matching gain is estimated:

$$\hat{g} = \sqrt{\frac{\|L_b\|_2^2}{\|R_b(\cdot + \hat{d})\|_2^2}}. \tag{10}$$

The gain is limited to ± 6 dB. Deviations smaller than 0.5 dB are mapped back to unity so that harmless stereo-energy differences do not spoil the exact $L-R$ solution. Finally, a smooth frequency weight $W[k]$ blends the compensated and basic outputs:

$$Y[k] = (1 - W[k])\, \mathcal{F}\{L - R\} + W[k], \mathcal{F}\{L - \hat{g}R(\cdot + \hat{d})\}. \tag{11}$$

The weight approaches one in the speech band and zero outside it. This limits the scope of the compensation and avoids an abrupt rectangular band edge.

4 Controlled experiment

4.1 Why a synthetic benchmark is necessary

With a commercial stereo song, the isolated vocal and instrumental stems are unknown. Waveform appearance, channel correlation, and listening impressions cannot measure vocal attenuation objectively. The benchmark therefore creates an eight-second stereo mixture from known components:

- a harmonic, amplitude-modulated vocal-like centre source;
- different tonal and filtered-noise accompaniment in each channel;
- a shared 75 Hz bass component;
- either ideal centring or a 0.82× gain and two-sample delay in the right-channel vocal.

Every learned model is estimated from the mixture and then applied unchanged to the known vocal and accompaniment components. Three metrics are reported: vocal leakage attenuation relative to the input vocal; scale-invariant accompaniment signal-to-distortion ratio (SDR); and the accompaniment-to-vocal energy ratio at the output. Values at ±80 dB indicate the numerical display floor or ceiling rather than infinite physical performance.

Table 1: Controlled benchmark results. Lower vocal attenuation is better; higher SDR and output ratio are better.

Scenario	Method	Vocal attn. (dB)	Acc. SDR (dB)	Output ratio (dB)
Ideal centre	$L - R$	−80.00	80.00	80.00
Ideal centre	Bass rescue	−61.50	18.44	60.26
Ideal centre	Adaptive	−80.00	80.00	80.00
Gain + delay mismatch	$L - R$	−10.69	80.00	10.27
Gain + delay mismatch	Bass rescue	−10.69	18.44	10.32
Gain + delay mismatch	Adaptive	−26.54	12.06	26.80

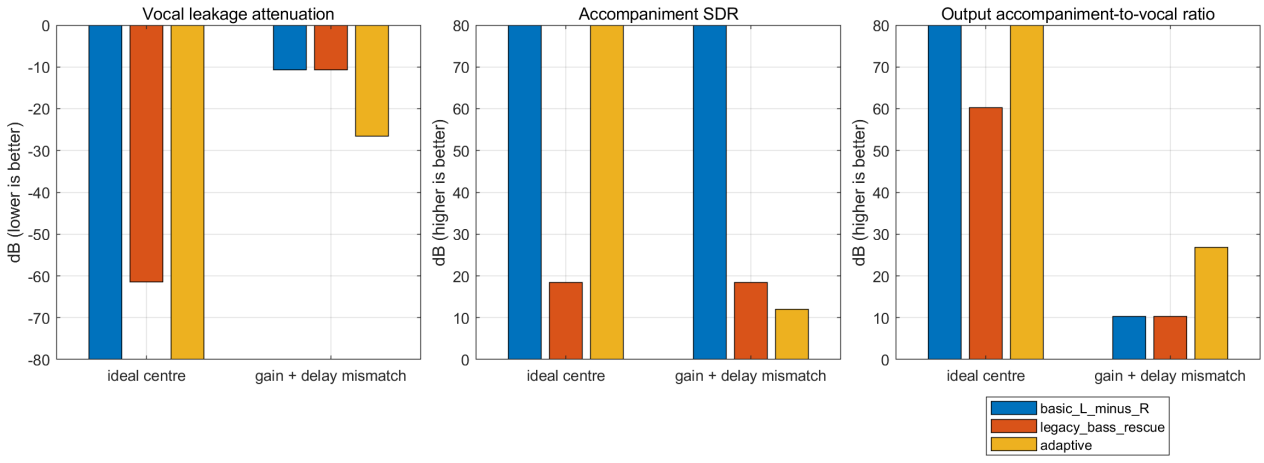


Figure 5: Benchmark summary. The compensated method substantially reduces vocal leakage under channel mismatch, but it also changes the accompaniment.

Under ideal centring, the simplest method is already optimal. Under mismatch, the adaptive method gains 15.85 dB of vocal attenuation and 16.54 dB in the output accompaniment-to-vocal ratio relative to $L - R$. Its 12.06 dB accompaniment SDR quantifies the cost. The result supports a conditional claim: gain-and-delay compensation is useful when the centre-source assumption is slightly violated, but it should not be enabled blindly.

5 Real-audio diagnostics

Two local stereo excerpts were processed for listening tests, but are excluded from version control. Because isolated stems are unavailable, Table 2 reports only descriptive stereo statistics. They are not labelled as vocal removal scores.

Table 2: Descriptive statistics for local test excerpts.

Case	F_s (Hz)	Duration (s)	L/R correlation	Mid/side energy (dB)
A	44100	35.016	0.906	13.065
B	44100	23.057	0.705	7.618

Case A has stronger centre coherence and is consequently a more favourable candidate for centre cancellation. Case B contains more stereo-side energy, so subtraction is less likely to isolate "instrumental only" content. Listening tests remain useful for detecting musical artifacts, but they do not replace the controlled metrics.

6 Limitations and reproducibility

- Centre cancellation cannot distinguish vocals from other centred sources. Modern stem-separation systems solve a different, harder problem.
- A single global delay and gain cannot model time-varying reverberation, chorus effects, nonlinear processing, or frequency-dependent panning.
- The synthetic benchmark is deliberately controlled and does not span the full diversity of commercial mixes.
- Project 1 time compression uses interpolation without an anti-aliasing prefilter, and its ideal DFT filter has circular-boundary artifacts.
- Real-song outputs are normalized with one common safety factor per song, rather than separately, so relative levels remain comparable.

The repository scripts use paths relative to each MATLAB file, create their own output directories, and generate figures non-interactively. The benchmark uses a fixed random seed. Source audio and generated WAV files are ignored by Git; the metrics CSV and figures are tracked.

7 Conclusion

The project now does more than demonstrate textbook operators. It shows where the assumptions behind centre cancellation succeed, where the original bass-rescue idea fails, and how a narrowly targeted correction changes both vocal leakage and accompaniment fidelity. The strongest portfolio value lies in this evidence-driven comparison: a simple baseline, a motivated improvement, an objective benchmark, and an explicit account of limitations.

References

[1] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed., Pearson, 2010.

[2] J. O. Smith III, *Spectral Audio Signal Processing*, W3K Publishing, 2011.

[3] Signals and Systems course project specification, internal course handout, 2026.